

# Introspective Imitation Dynamics and the Limits of Extrinsic Update Rules in Heterogeneous Public Goods Games

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May 8, 2026

## Abstract

Evolutionary game theory provides a framework by which to study the emergence of cooperation in a population of self-interested actors. In such a framework, players’ decision on whether or not to cooperate evolve according to decision rules called population dynamics. However, often games are studied under the assumption that all individuals play under the same conditions. In this paper, we categorise and compare four different population dynamics as “extrinsic”, where players learn by looking outward at the other players in the population, and “intrinsic”, where players look inwardly at their own attributes or potential payoffs. We show that extrinsic population dynamics have a ceiling on the rate of cooperation which can be exceeded by intrinsic population dynamics, and demonstrate this using the public goods game with heterogeneous contributions. We then introduce “introspective imitation dynamics”, a novel population dynamic which combines both fitness-proportional extrinsic selection and self-referential intrinsic update steps, which overcomes the inherent limitations of the other four population dynamics.

## 1 Introduction

Reducing global carbon emissions is one of the defining collective action problems of our time. Countries differ substantially in their economic capacity, historical emissions, and the costs they face when decarbonising, yet they share a mutual interest in the global public good of a stable climate. However, they face the temptation to free-ride, accepting the benefits of the public good without contributing to the cost [30]. International agreements have attempted to encourage a reduction in global emissions, for example the Kyoto protocol [1] and Paris agreement [4, 16]. Most recently, the European Union’s Carbon Border Adjustment Mechanism [8], which requires importers to purchase certificates reflecting the carbon content of goods produced under weaker environmental regulation, represents a real-world instance of this idea.

This structure maps onto a *public goods game* [39, 15, 2, 14, 13]: each actor decides whether to incur a private cost to contribute to a shared resource whose total value is multiplied by some factor  $r$  and distributed equally. The temptation to free-ride means that purely selfish reasoning leads to a tragedy of the commons [12, 47], even when collective cooperation would benefit everyone. Both theory [37] and controlled experiments [28, 27] show that framing cooperation as a collective-risk problem, where failure to meet a cooperation threshold incurs a shared loss, can substantially raise cooperation rates. The challenge of overcoming free-riding is not merely theoretical. Nordhaus [30] proposed voluntary *climate clubs* in which member nations adopt a common carbon price and levy tariffs on non-members, turning the international climate problem into a coordination game with enforcement.

Evolutionary game theory offers a framework for studying how cooperation can emerge among self-interested actors without assuming rational foresight [17, 31]. In this approach, strategies that perform well spread through a population via one of several update rules, while poorly performing strategies decline. The conditions under which cooperative strategies can invade and stabilise have been extensively studied in two-player settings such as the iterated prisoner’s dilemma [21, 3] and the snowdrift game [46]. The public goods game generalises these pairwise dilemmas to  $N$  players [13], making it well suited to studying multi-lateral cooperation such as climate agreements. In finite populations, evolutionary outcomes are stochastic: a rare mutant strategy spreads or vanishes with a probability that depends on the population size, payoff structure, and update rule [33, 45].

A feature of real-world collective action that standard models typically set aside is that actors are not identical. Players differ in wealth, productive capacity, and the costs they face, and this inequality can reshape the dynamics of cooperation [15]. In spatial and graphical settings, where players sit on a network and participate simultaneously in multiple games with their neighbours [32, 42, 34], heterogeneity generally promotes cooperation. In [38] it is shown that social diversity in endowments promotes the emergence of cooperation in the public goods game directly: heterogeneous populations outperform their homogeneous counterparts over a wide range of parameters. This echoes earlier work showing that scale-free network topology, a form of structural heterogeneity, provides a unifying framework for the emergence of

cooperation [36]. When contributions depend on dynamically accumulated reputation [26], high-reputation defectors receive reduced transfers, weakening defection’s advantage. When heterogeneity arises from differences in network degree so that players allocate contributions according to the strength of their social ties [23], the payoff structure penalises high-degree defectors and protects cooperating clusters from invasion. The characteristic spatial mechanism is the formation of cooperating clusters that are more profitable than their defecting neighbours; provided  $r$  is sufficiently large, these clusters resist invasion and eventually spread. Heterogeneity does not, however, unconditionally favour cooperation. Flores et al. [9] show that when the contribution level is itself a strategic variable rather than a fixed attribute, low-value contributions act as a form of defection against the most collectively beneficial strategies, so that the emergence of cooperation does not guarantee the emergence of high-value cooperation.

The studies above are almost exclusively concerned with spatially structured populations, where network topology is central. Much less is known about fully heterogeneous, *well-mixed* populations, in which all individuals are distinguished by a fixed attribute (such as their contribution capacity) independently of any graph structure. Furthermore, most existing analyses rely on simulation or mean-field approximations; exact treatments of heterogeneous evolutionary dynamics remain rare. How the choice of update rule interacts with population heterogeneity is also poorly understood. The Moran process [29, 20], Fermi imitation dynamics [43], aspiration dynamics [7], and introspection dynamics [5, 6] all model strategy revision in distinct ways, and each may respond differently to player-specific payoffs. In particular, *intrinsic* dynamics, in which a player evaluates a prospective strategy change based on their own counterfactual payoff, are better suited to heterogeneous populations than *extrinsic* dynamics, in which a player may copy a high-fitness neighbour without considering whether the same strategy would serve them as well.

We make the following contributions. First, we build on an established a general framework for evolutionary dynamics on fully heterogeneous, well-mixed populations of  $N$  ordered individuals, in which each player may have a distinct payoff function and contribution level. Second, we formalize the notion of a purely extrinsic population dynamic, a definition encompassing both the Moran process [29, 20] and Fermi imitation dynamics [43], and establish that such dynamics are inherently constrained by an upper bound on the probability of cooperation. Third, we consider two intrinsic population dynamics, Aspiration dynamics [7] and Introspection dynamics [5, 6], show that these processes can overcome the ceiling of purely extrinsic dynamics, and discuss their weaknesses. Finally, we introduce a novel update rule, *Introspective Imitation Dynamics*, in which players select a role model proportionally to fitness (as in the Moran process) but evaluate whether to adopt that strategy by comparing their own current and counterfactual payoffs (as in introspection dynamics). This rule is well suited to heterogeneous populations, where copying a high-fitness individual is likely although irrational if the same strategy yields a worse payoff for the copying player. For each population dynamic, we will use the public goods game [39, 15, 2, 14, 13] to demonstrate our results.

The remainder of the paper is organised as follows. Section 2 introduces the population model and the public goods game. Section 3 introduces a formal definition of a *purely extrinsic population dynamic*, gives formulations of the Moran process and Fermi imitation dynamics in terms of our population model, proves that these satisfy the definition of a purely extrinsic population dynamic, and proves the ceiling on the probability of cooperation in such dynamics. We also demonstrate this in a public goods game. Section 3 gives formulations of two intrinsic population dynamics and shows that they can exceed the ceiling on the probability of cooperation. We also discuss their disadvantages, and show their results in the public goods game. Section 4 introduces *introspective imitation dynamics*, a novel population dynamic, and show that it exhibits none of the disadvantages present in the other 4 dynamics. We then show under which conditions introspective imitation dynamics is able to exceed the cooperation ceiling in the public goods game.

## 2 The General Heterogeneous Population Dynamic Model

Following [5, 6], we define a general framework for evolutionary dynamics on a fully heterogeneous, well-mixed population.

### 2.1 Population and State Space

The population consists of  $N$  **ordered** individuals. Each individual selects an action from a common action set  $A = (a_1, a_2, \dots, a_k)$ ; we assume throughout that all individuals share the same action set of size  $k$  although this is not strictly required for introspection and aspiration dynamics [5, 7]. A *state* of the population is an ordered  $N$ -tuple  $\mathbf{a} = (a_1, a_2, \dots, a_N) \in A^N$ , and the *state space*  $S$  is the set of all such tuples. A payoff function  $\pi: S \rightarrow \mathbb{R}^N$  assigns to each state a vector of individual payoffs; we write  $\pi_i(\mathbf{a})$  for the payoff of individual  $i$  in state  $\mathbf{a}$ .

The population evolves as a Markov chain [40] on  $S$ . Transitions can only occur between states that differ in exactly one coordinate, that is, states  $\mathbf{a}$  and  $\mathbf{b}$  with Hamming distance  $h(\mathbf{a}, \mathbf{b}) = 1$  [22]. We call the unique coordinate at which  $\mathbf{a}$  and  $\mathbf{b}$  differ the *index of difference*, denoted  $I(\mathbf{a}, \mathbf{b})$  [6]. The set of states reachable from  $\mathbf{a}$  in one step is  $\text{Neb}(\mathbf{a}) = \{\mathbf{b} \in S : h(\mathbf{a}, \mathbf{b}) = 1\}$  [6].

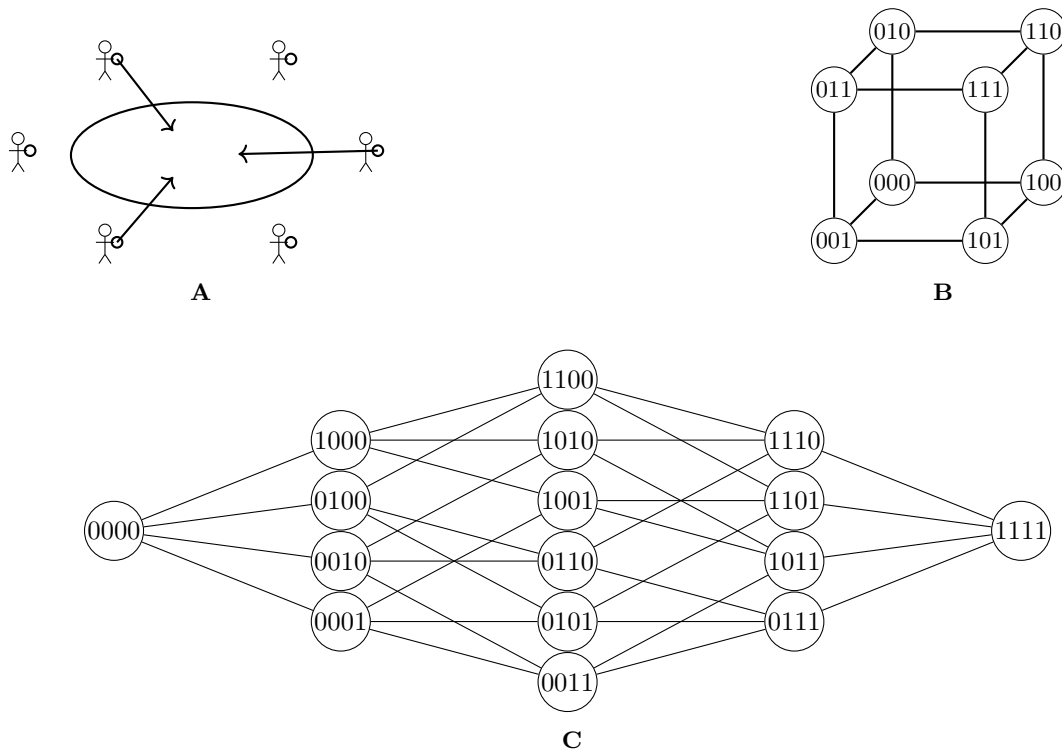


Figure 1: **A**: The public goods game with 6 players. Here 3 players contribute. **B**: The graph of a 3-player game with 2 actions under a birth-death process. This is closely related to the hypercube of dimension 3. **C**: The graph of a 4-player game with 2 actions under a birth-death process. This representation shows how one player changes action at a given time.

If we consider a game with two actions, for example cooperate and defect, we can compare the graph of such a Markov chain to an  $N$ -dimensional hypercube. This is the graph whose node set consists of the  $2^N$   $N$ -dimensional boolean vectors, and which has edges between edges if and only if they differ in exactly one position [11]. The state space of an evolutionary game with two actions can be modelled in this way by denoting the strategies 0 and 1.

The graphical representation of a birth-death process differs from a hypercube in exactly two ways. First, a population may remain the same at a given step, and thus each vertex contains a self-loop. Additionally, the probability of transitioning from a state  $\mathbf{a}$  to a state  $\mathbf{b}$  is not necessarily the same as the probability of transitioning from  $\mathbf{b}$  to  $\mathbf{a}$ . Thus, instead of one edge between two states, we have two directed edges, one in each direction. Below we see a comparison of the 3-dimensional hypercube and the graph of a 3 player game with 2 actions.

## 2.2 Mutation

Three of the dynamics that we discuss in this paper correspond to absorbing Markov chains, where the only relevant outcome is the probability of absorption into a given absorbing state. The remaining two dynamics are ergodic chains, for which we can compute the long-run probability of cooperation. To enable meaningful comparison across all models, we introduce mutation into our framework.

Mutation ensures that all processes become ergodic, while also introducing a controlled amount of noise: at any moment, individuals may switch to a different strategy. [10]

Given an original process and two states  $\mathbf{a}$  and  $\mathbf{b}$ , the probability of the original process *with mutation* transitioning from  $\mathbf{a}$  to  $\mathbf{b}$  is given by:

$$P(\mathbf{a} \rightarrow \mathbf{b}) = P(\text{pick individual } I(\mathbf{a}, \mathbf{b})) \cdot (P(\mathbf{a} \rightarrow \mathbf{b}) \cdot P(\text{no mutation}) + P(\mathbf{a}_{I(\mathbf{a}, \mathbf{b})} \text{ mutates} \rightarrow \mathbf{b}_{I(\mathbf{a}, \mathbf{b})}))$$

We assume a mutation parameter  $\mu \in [0, 1]_{\mathbb{R}}^{N \times k}$ , where  $\mu_{ij}$  denotes the probability that individual  $i$  adopts strategy  $j$  via mutation. For a population process with transition matrix  $T$ , the mutated process is described by  $T^{(\mu)}$ . In this process,

a transition from state  $\mathbf{a}$  to  $\mathbf{b}$  occurs with probability  $T_{\mathbf{ab}}$ ; mutation then acts on the resulting state with probability dictated by  $\mu$ .

For a focal individual  $i$ , the probability that no mutation occurs is  $1 - \sum_{j=1}^k \mu_{ij}$ . Let  $T_{\mathbf{ab}}$  be the transition matrix for a population dynamic. Then the transition matrix  $T^{(\mu)}$  for the process with mutation is:

$$T_{\mathbf{ab}}^{(\mu)} = \begin{cases} T_{\mathbf{ab}} \left(1 - \sum_{j=1}^k \mu_{I(\mathbf{a}, \mathbf{b}), j}\right) + \frac{\mu_{I(\mathbf{a}, \mathbf{b}), I(\mathbf{a}, \mathbf{b})}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \quad (1)$$

Note that if  $\sum_{j=1}^k \mu_{ij} = 1$ , mutation fully determines the behavior of individual  $i$ .

This construction ensures that the process is ergodic, rather than absorbing. Consequently, we no longer obtain absorption matrices for models with mutation; instead, we derive a unique steady state that is independent of the initial state. All processes in this paper are considered under the effect of homogeneous mutation, where  $\mu_{ik} = \mu_{il}$  for all action types  $k, l$ .

## Measures of Cooperation

We define  $p_C$  to be the *probability of cooperation*. This represents the mean proportion of cooperators at a given step once an evolutionary game reaches its steady state, and can be calculated by the value  $\frac{\sum_{\mathbf{a} \in S} (\prod_a \sum_{i=1}^N \mathbf{a}_i)}{N}$ , where  $\mathbf{a}_i$  takes the value 1 if player  $i$  cooperates in state  $\mathbf{a}$ , and 0 if not. A special case of  $p_C$  is in an extrinsic population dynamic where  $\mu = 0$ . In such a case, our dynamic forms an absorbing Markov chain, and as such rather than a steady state we obtain an absorption matrix which depends on the initial state of the system. In this case,  $p_C$  can be interpreted as the probability that, given an initial state, the chain will be absorbed into the state where all players cooperate. Figure ?? shows the marginal distribution of the cooperation probability  $p_C$  across all parameter combinations and dynamics. Most configurations yield low cooperation, but the right tail is non-trivial: meaningful cooperation is achievable under a subset of conditions. We also see the distribution of  $p_C$  across each population dynamic.

## 2.3 The Public Goods Game

To illustrate all of the dynamics of Section 2 we will use the Public Goods game.

We apply the dynamics above to the public goods game [39, 15, 2, 14, 13]. To illustrate the results in the following sections, we use the public goods game. [39, 15, 2, 14, 13]. In this game, individuals choose whether or not to contribute to a common resource pool. The total contribution is multiplied by a factor  $r > 1$  and distributed equally among all  $N$  players. The model captures the temptation to free-ride [12]: a player who withholds their contribution still receives an equal share of the multiplied pool, so individual incentives push against the collectively optimal outcome of universal cooperation.

We can model a public goods game by providing the following payoff function:

$$\pi_i(\mathbf{a}) = \frac{r \sum_{j=1}^N \alpha_j}{N} - \alpha_i \quad (2)$$

Where  $\alpha_j$  is the contribution by individual  $j$ . This is a similar heterogeneous public good game as the one described in [15] although in that model the return  $r$  is also a heterogeneous property of each player, whereas we only consider heterogeneity in contributions. We consider a “linear” population, in which a player in position  $i$  contributes  $\frac{2Mi}{N(N+1)}$ , where  $M$  is the total amount contributed when all players play  $C$ . In such a case, the payoff of all players is  $\pi_i = \frac{rM}{N} - \alpha_i$ . Supplementary figure 1 shows the combinations of parameters considered across each of the population dynamics.

It can be seen that full contribution is the Pareto optimal state, however if  $r < N$ , full defection forms the Nash equilibrium for the one shot game. When  $r > N$ , i.e the value of the public good is multiplied by a greater amount than the number of players, each player will profit from their own contribution, as we have the value  $\Delta(\pi_i) = \pi_C(\mathbf{a}) - \pi_D(\mathbf{b}) = \alpha_i(\frac{r}{N} - 1) > 0$  if and only if  $r > N$ . Therefore, in the case  $r > N$ , the Nash equilibrium of the one-shot game is full contribution. However, it is important to note that in a given mixed state in the evolutionary model of cooperation in the public goods game, a defector will have a higher payoff than a cooperator, no matter the value of  $r$ . This is because all players receive the same return from the public good, whilst only cooperators incur a cost  $-\alpha_i$ .

To consider the distribution of the probability of cooperation in a public goods game, a large scale parameter sweep is carried out. To the authors knowledge this is the largest such parameter sweep. It is done to the highest standards of reproducibility [48, 35, 49, 41, 44] and all data has been archived and is a further contribution of the work. For the

purpose of this research, the authors developed a software library `ludics` (written in python) which is installable. The archive of the software is at [?] and the documentation can be found at <https://hefos.github.io/ludics/>. All the development of the software can be found at <https://github.com/hefos/ludics>.

### 3 Extrinsic population dynamics

We begin by defining an *extrinsic population dynamic*. Intuitively, a purely extrinsic process revises strategies through payoff comparison between members of a population.

**Definition 1** (Extrinsic population dynamic). *Let  $T$  be the transition matrix of a population dynamic on a state space  $S$ . We say that  $T$  defines a purely extrinsic process if for every  $\mathbf{a}, \mathbf{b} \in S$  with  $\mathbf{b} \in \text{Neb}(\mathbf{a})$ ,  $T_{\mathbf{ab}}$  satisfies:*

1.  $T_{\mathbf{ab}}$  is a function of payoffs evaluated in  $\mathbf{a}$  and in no other state.
2. Let  $\mathcal{R}$  denote the set of players  $j$  such that  $j \neq I(\mathbf{a}, \mathbf{b})$  and  $\mathbf{a}_j = \mathbf{b}_{I(\mathbf{a}, \mathbf{b})}$ . Then  $T_{\mathbf{ab}}$  is non-decreasing in the payoffs  $\pi_j(\mathbf{a})$  for all  $j \in \mathcal{R}$  (and strictly increasing in at least one such  $j$  if  $\mathcal{R} \neq \emptyset$ ), and non-increasing in the payoffs  $\pi_j(\mathbf{a})$  for all  $j \notin \mathcal{R}$ .
3.  $T_{\mathbf{ab}}$  contains no term dependent on  $a_{I(\mathbf{a}, \mathbf{b})}$  or  $b_{I(\mathbf{a}, \mathbf{b})}$ , except in the payoff function  $\pi_{I(\mathbf{a}, \mathbf{b})}(\mathbf{a})$ , mutation parameters  $\mu_{ij}$ , and in the definition of the set  $\mathcal{R}$ .
4. If all players share the same payoff in state  $\mathbf{a}$  (that is, there exists  $v \in \mathbb{R}$  such that  $\pi_i(\mathbf{a}) = v$  for every player  $i$ ), then  $T_{\mathbf{ab}}$  is independent of the common value  $v$ .
5. Let  $T^{\text{nd}}$  denote  $T$  with all payoffs set to a common value, well-defined by (4). Order  $\{C, D\}^N$  componentwise with  $D < C$ . For every  $\mathbf{a} \leq \mathbf{a}'$ , there exist random variables  $\mathbf{X}, \mathbf{X}'$  on a common probability space, with  $\mathbf{X} \sim T^{\text{nd}}(\mathbf{a}, \cdot)$  and  $\mathbf{X}' \sim T^{\text{nd}}(\mathbf{a}', \cdot)$ , such that  $\mathbf{X} \leq \mathbf{X}'$  almost surely.

Condition (2) captures the basic monotonicity: a higher payoff for a role model makes that role model more likely to be copied, and a higher payoff for a non-role-model pulls the other way. Condition (3) rules out an intrinsic preference for either action. Conditions (4) and (5) work as a pair, ruling out hidden biases that survive when no payoff comparison is available.

Condition (4) captures payoff-scale neutrality: when all players earn the same payoff, the absolute level of that common value cannot influence  $T_{\mathbf{ab}}$ . Condition (5) captures configuration neutrality: under those same neutral payoffs, starting from a more cooperative profile cannot make the next state any less cooperative. Put simply: at neutral payoffs, a population that starts at least as cooperative as another remains at least as cooperative one step later. Without (5), a process could satisfy (1)–(4) and still embed a configuration-dependent pull toward defection at neutral payoffs, which is an intrinsic bias rather than truly neutral behaviour. [19] notes that birth-death processes are stochastically monotone, and we shall show that the Moran process and Fermi imitation dynamics are also stochastically monotone.

*Note:* Condition (3) in the above definition serves to exclude population dynamics which explicitly favour a certain action type, for example:

$$\bar{T}_{\mathbf{ab}} = \begin{cases} \delta_{I(\mathbf{a}, \mathbf{b})} \cdot T_{\mathbf{ab}} & \text{if } \mathbf{b} \neq \mathbf{a}, \\ 1 - \sum_{c \neq \mathbf{a}} T_{\mathbf{ac}} & \text{if } \mathbf{b} = \mathbf{a}. \end{cases}$$

where  $T_{\mathbf{ab}}$  is the transition matrix of the some purely extrinsic dynamic, and  $\delta_{I(\mathbf{a}, \mathbf{b})}$  takes the value 1 if  $b_{I(\mathbf{a}, \mathbf{b})} = C$  and  $\frac{1}{2}$  if not. This population dynamic would be a purely extrinsic population dynamic, but would be tailored to favour a particular action type. We exclude cases such as this, as they do not conform to the framework in which fitness is the single measure of the success of action types in the population, and thus of which strategy is passed on in each step of the process. While parameters such as choice intensity and selection intensity affect the transition probability  $T_{\mathbf{ab}}$ , they do not inherently favour any specific action type.

We now show two classical examples of purely extrinsic population dynamics

#### 3.1 The Moran Process

The Moran process [29] is defined by the following algorithm:

1. A player is chosen to be copied with a probability proportional to their fitness in the population

2. A player is chosen with probability  $\frac{1}{N}$  to change their strategy

The process goes from state to state according to the entry  $T_{\mathbf{ab}}$  of a transition matrix  $T$  in which each row and column correspond to a state in  $S$  is defined as the probability of a transition  $\mathbf{a} = (a_1, a_2, \dots, a_N) \rightarrow \mathbf{b} = (b_1, b_2, \dots, b_N)$ . For the Moran process, this is as follows:

$$T_{\mathbf{ab}} = \begin{cases} \left( \frac{\sum_{a_i=b_{I(\mathbf{a},\mathbf{b})}} f_i(\mathbf{a})}{N \sum_{a_j} f_j(\mathbf{a})} \right) \left( 1 - \sum_{j=1}^k \mu_{I(\mathbf{a},\mathbf{b}),j} \right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \text{ differing at position } I(\mathbf{a},\mathbf{b}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (3)$$

Where  $f_i(\mathbf{a}) = 1 + \epsilon \pi_i(\mathbf{a})$  is the fitness of a player  $i$  in state  $\mathbf{a}$  scaled with selection intensity  $\epsilon$ . This ensures a positive fitness for all players in order that the probability  $T_{\mathbf{ab}}$  is always well defined. The selection intensity  $\epsilon$  controls the rationality with which players choose which strategy to adopt in each time step - a larger  $\epsilon$  indicates that a player will choose a higher fitness strategy with a greater probability. This value of  $\epsilon$  must, however, be chosen in order that  $f_i(\mathbf{a})$  is strictly positive.

The Moran process is classically defined for natural evolution, where a player's "type" represents a mutation within a population. In each time step, rather than a given player choosing a new action type, a randomly chosen individual in the population would die and be removed from the population. Another individual would reproduce with a probability proportional to their fitness, introducing an exact copy of themselves to the population to replace the removed individual. As a result of the model's original purpose, the Moran process contains no decision step where a player chooses whether or not to accept a considered strategy, as in natural selection a player cannot choose their . Instead, at each time step, a player *must* accept another player's action, and we remain in the same state when a player accepts the same action as they currently play.

One implication of this is that in a heterogeneous population, worsening moves may be taken with a high probability. This is because an action that provides a high payoff for one player in a population may not provide the same return for another. For example, in the public goods game with heterogeneous returns, a player with a low  $r_i < N$  may copy a player who contributes with a high  $r_j > N$  due to their high individual payoff. However, this would be a worsening move for the focal player  $i$ , as due to their low individual rate of return they would lower their payoff.

We now show that the Moran process satisfies the conditions of 1

**Theorem 1** (The Moran process is a purely extrinsic process).

*Proof.* By definition the Moran process satisfies conditions (1) and (3). It remains to be shown that  $T_{\mathbf{ab}}$  is non-decreasing in  $\pi_j(\mathbf{a})$  for  $j \in \mathcal{R}$  (strictly increasing for at least one such  $j$ ) and non-increasing in  $\pi_j(\mathbf{a})$  for  $j \notin \mathcal{R}$ . Recall  $f_i(\mathbf{a}) = 1 + \epsilon \pi_i(\mathbf{a})$  for some selection intensity  $\epsilon > 0$ . Then we have:

$$T_{\mathbf{ab}} = \frac{\sum_{j \in \mathcal{R}} f_j(\mathbf{a})}{N \sum_{l=1}^N f_l(\mathbf{a})} \left( 1 - \sum_{i=1}^k \mu_{I(\mathbf{a},\mathbf{b}),i} \right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

$$\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = \frac{\epsilon \sum_{x \notin \mathcal{R}} f_x(\mathbf{a})}{N (\sum_{l=1}^N f_l(\mathbf{a}))^2} \left( 1 - \sum_{i=1}^k \mu_{I(\mathbf{a},\mathbf{b}),i} \right) \quad \text{for } j \in \mathcal{R}$$

For  $j \in \mathcal{R}$ , as  $\epsilon$  is chosen such that  $f_i(\mathbf{a}) > 0$  for all  $i \in \mathbf{a}$ , and  $\mu_{i,j}$  such that  $\sum_{i=1}^k \mu_{I(\mathbf{a},\mathbf{b}),i} < 1$ , this derivative is strictly positive. For  $j \notin \mathcal{R}$ , the analogous computation gives  $\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = -\frac{\epsilon \sum_{x \in \mathcal{R}} f_x(\mathbf{a})}{N (\sum_{l=1}^N f_l(\mathbf{a}))^2} (1 - \sum_{i=1}^k \mu_{I(\mathbf{a},\mathbf{b}),i}) \leq 0$ , so condition (2) holds. Finally, for condition (4), setting  $\pi_i(\mathbf{a}) = v$  for every player  $i$  gives  $f_i(\mathbf{a}) = 1 + \epsilon v$  for every  $i$ , whence  $T_{\mathbf{ab}} = \frac{|\mathcal{R}|(1+\epsilon v)}{N \cdot N(1+\epsilon v)} (1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} = \frac{|\mathcal{R}|}{N^2} (1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$ , which is independent of  $v$ .

For condition (5), specialise to two actions  $\{C, D\}$  with  $D < C$ . Under  $T^{\text{nd}}$ , the dynamic samples a focal player  $i$  uniformly, a role model  $k$  uniformly from the population (since all fitnesses are equal), and a mutation outcome  $m$  from the action-invariant mutation distribution; player  $i$  then adopts  $a_k$ , or  $m$  if a mutation occurs. Couple the chains started at  $\mathbf{a} \leq \mathbf{a}'$  on the common probability space generated by  $(i, k, m)$ , applying the same draw to both. Coordinate  $i$  in the successor of  $\mathbf{a}$  is  $a_k$  (or  $m$ ); coordinate  $i$  in the successor of  $\mathbf{a}'$  is  $a'_k$  (or the same  $m$ ). Since  $\mathbf{a} \leq \mathbf{a}'$  gives  $a_k \leq a'_k$ , and every other coordinate  $j$  is left unchanged with  $a_j \leq a'_j$ , the two successors  $\mathbf{X}$  and  $\mathbf{X}'$  satisfy  $\mathbf{X} \leq \mathbf{X}'$  pointwise on this probability space, so condition (5) holds.  $\square$

### 3.2 Fermi Imitation Dynamics

The Fermi Imitation Dynamic is another method of modelling evolution in a population. It models the following process:

- Two players,  $i$  and  $j$  are selected at random. Unlike the original formulation [43], player  $j$  need not be a neighbour of player  $i$ , reflecting the well-mixed setting considered here.
- Player  $i$  copies the action type of the player  $j$  with a probability according to the Fermi function  $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1 + e^{\beta(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a}))}}$

Where  $\beta$  is the choice intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff, and the payoff which they are considering. This method has been commonly used in studies of evolutionary games, for example [26, 23, 43]. We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} \left( \frac{1}{N(N-1)} \sum_{a_j=b_{I(\mathbf{a}, \mathbf{b})}} \phi(\pi_{I(\mathbf{a}, \mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})) \right) \left( 1 - \sum_{j=1}^k \mu_{I(\mathbf{a}, \mathbf{b}), j} \right) + \frac{\mu_{I(\mathbf{a}, \mathbf{b}), b_{I(\mathbf{a}, \mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (4)$$

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. Instead, Fermi Imitation Dynamics compares the fitness after the selection of players. This gives the comparison step in which a player may choose not to adopt a new strategy if it admits a lower fitness in the current population. However, equivalently to the Moran process, in a heterogeneous population the method of adopting a new strategy based entirely on the payoff of another individual may lead to worsening moves in which players adopt strategies which currently admit a higher fitness, but which will not improve their own fitness.

We also see that due to the formulation of the logit function  $\phi(\Delta(\pi))$ , Fermi imitation dynamics can become invariant under certain parameters. Take, for example, the public goods game. When comparing a cooperator and defector under  $\phi$ , we have the value  $\Delta(\pi) = \pm \alpha_i$ , and thus we see that the parameter  $r$  has no effect on the probability of accepting a new strategy under this population dynamic. This occurs in any game in which one action type's payoff in a given state can be expressed as another action type's payoff up to the addition of some factor  $\gamma$ , where  $\gamma$  depends only on a subset of the game's parameters. In that case,  $\Delta(\pi) = \pm \gamma$ , and is therefore invariant to any parameters that do not influence  $\gamma$ .

We now show that Fermi imitation dynamics satisfies the definition of a purely extrinsic population dynamic

**Theorem 2** (Fermi imitation dynamics is a purely extrinsic population dynamic).

*Proof.* By definition, Fermi imitation dynamics satisfies conditions (1) and (3). It remains to show that it satisfies conditions (2), (4), and (5).

$$T_{\mathbf{ab}} = \frac{1}{N(N-1)} \sum_{j \in \mathcal{R}} \phi(\pi_{I(\mathbf{a}, \mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})) \left( 1 - \sum_{i=1}^k \mu_{I(\mathbf{a}, \mathbf{b}), i} \right) + \frac{\mu_{I(\mathbf{a}, \mathbf{b}), b_{I(\mathbf{a}, \mathbf{b})}}}{N}$$

As  $\phi$  is strictly decreasing, every term in the sum is strictly increasing in  $\pi_j(\mathbf{a})$  for  $j \in \mathcal{R}$ . For  $j \notin \mathcal{R}$  with  $j \neq I(\mathbf{a}, \mathbf{b})$ ,  $\pi_j(\mathbf{a})$  does not appear in the formula at all. For  $j = I(\mathbf{a}, \mathbf{b})$  (when  $a_{I(\mathbf{a}, \mathbf{b})} \neq b_{I(\mathbf{a}, \mathbf{b})}$ , so  $I \notin \mathcal{R}$ ), each term is strictly decreasing in  $\pi_{I(\mathbf{a}, \mathbf{b})}(\mathbf{a})$  because  $\phi$  is. Thus, Fermi imitation dynamics satisfies condition (2). Finally, condition (4) holds: if  $\pi_i(\mathbf{a}) = v$  for every player  $i$ , then every payoff difference appearing in the Fermi formula is zero, so  $\phi(\pi_{I(\mathbf{a}, \mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})) = \phi(0) = \frac{1}{2}$  in each summand, and  $T_{\mathbf{ab}} = \frac{|\mathcal{R}|}{2N(N-1)} \left( 1 - \sum_i \mu_{I(\mathbf{a}, \mathbf{b}), i} \right) + \frac{\mu_{I(\mathbf{a}, \mathbf{b}), b_{I(\mathbf{a}, \mathbf{b})}}}{N}$ , independent of  $v$ .

For condition (5), specialise to two actions  $\{C, D\}$  with  $D < C$ . Under  $T^{\text{nd}}$ , the dynamic samples a focal player  $i$  uniformly, a role model  $k$  uniformly from the remaining  $N-1$  players, an independent Bernoulli outcome with success probability  $\phi(0) = \frac{1}{2}$ , and a mutation outcome  $m$  from the action-invariant mutation distribution; player  $i$  adopts  $a_k$  on a success (or  $m$  if a mutation occurs) and otherwise keeps  $a_i$ . Couple the chains started at  $\mathbf{a} \leq \mathbf{a}'$  on the common probability space generated by  $(i, k, \text{Bernoulli}, m)$ , applying the same draw to both. On a mutation step, both successors set coordinate  $i$  to the same  $m$ . On a successful imitation step, they set coordinate  $i$  to  $a_k$  and  $a'_k$  respectively, with  $a_k \leq a'_k$ . On a no-update step, coordinate  $i$  retains  $a_i \leq a'_i$ . Every other coordinate is unchanged, so the two successors  $\mathbf{X}$  and  $\mathbf{X}'$  satisfy  $\mathbf{X} \leq \mathbf{X}'$  pointwise on this probability space, and condition (5) holds.  $\square$

### 3.3 The Cooperation Ceiling under Purely Extrinsic Population Dynamics

We will now show that there is a theoretic limit on the probability of cooperation  $p_C$  for any purely extrinsic population dynamic. Before stating the theorem we record a comparison result for Markov chains on partially ordered spaces. The result is classical and we claim no novelty; we include a short proof for the reader's convenience, since its use in the proof of the theorem below may not be familiar to the reader.

**Lemma 1.** *Let  $P$  and  $Q$  be irreducible, aperiodic transition matrices on a finite partially ordered set  $(S, \leq)$ , with stationary distributions  $\pi^P$  and  $\pi^Q$ . Suppose that*

- (i)  $P(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}, \cdot)$  for every  $\mathbf{a} \in S$ ; and
- (ii)  $Q$  is stochastically monotone:  $\mathbf{a} \leq \mathbf{a}'$  implies  $Q(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}', \cdot)$ .

Then  $\pi^P \preceq_{st} \pi^Q$ .

*Proof.* Kamae, Krengel, and O'Brien [18] prove that whenever two probability measures  $\mu_1, \mu_2$  on a partially ordered Polish space satisfy  $\mu_1 \preceq_{st} \mu_2$ , there exist random variables  $X_1 \sim \mu_1$  and  $X_2 \sim \mu_2$  defined on a common probability space with  $X_1 \leq X_2$  almost surely (a *monotone coupling*).

Using this, we construct a coupling of the two chains by induction on  $t$ . Start with any  $X_0^P, X_0^Q \in S$  satisfying  $X_0^P \leq X_0^Q$ . Given  $X_t^P \leq X_t^Q$ , hypothesis (ii) gives  $Q(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$ , which combined with (i) at  $\mathbf{a} = X_t^P$  yields  $P(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$ . Applying Kamae–Krengel–O'Brien to this pair of measures produces  $(X_{t+1}^P, X_{t+1}^Q)$  with the required marginals and  $X_{t+1}^P \leq X_{t+1}^Q$  almost surely. By induction,  $X_t^P \leq X_t^Q$  for every  $t$ . Since both chains are ergodic, their marginal laws converge to  $\pi^P$  and  $\pi^Q$ , and passing to the limit gives  $\pi^P \preceq_{st} \pi^Q$ .  $\square$

**Theorem 3** (Extrinsic Ceiling). *Let  $T$  define a purely extrinsic process on an  $N$ -player game with 2 actions,  $C$  and  $D$ , such that in any given state  $\mathbf{a}$ ,  $\min_{j:a_j=D} f_j(\mathbf{a}) > \max_{i:a_i=C} f_i(\mathbf{a})$  (that is, in any given state, any defector has greater fitness than any cooperator). Suppose further that the system exhibits action-invariant mutation  $\mu_{iC} = \mu_{iD}$  for all players  $i \in \mathbf{a}$ .*

*Then,  $p_C \leq \frac{1}{2}$*

*Proof.* Equip  $S = \{C, D\}^N$  with the componentwise partial order, declaring  $D < C$ . For a transition matrix  $P$ , write  $P(\mathbf{a}, \cdot) \preceq_{st} P(\mathbf{a}', \cdot)$  to mean that the one-step distribution from  $\mathbf{a}$  under  $P$  is stochastically dominated by the one-step distribution from  $\mathbf{a}'$  under  $P$  in this order. Let  $T_{\mathbf{ab}}^{\text{nd}}$  denote the value of  $T_{\mathbf{ab}}$  evaluated at  $\pi_i(\mathbf{a}) = 0$  for every  $i$ ; by condition (4),  $T_{\mathbf{ab}}^{\text{nd}}$  is equally the value of  $T_{\mathbf{ab}}$  under any other choice of common payoff.

*Step 1:*  $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$  for every  $\mathbf{a} \in S$ .

Set  $v(\mathbf{a}) := \min_{a_j=D} \pi_j(\mathbf{a})$ . In every mixed state  $\mathbf{a}$ , the payoff-dominance hypothesis implies  $\pi_i(\mathbf{a}) < v(\mathbf{a})$  for every cooperator  $i$  and  $\pi_j(\mathbf{a}) \geq v(\mathbf{a})$  for every defector  $j$ . Consider a  $C$ -gaining transition  $\mathbf{a} \rightarrow \mathbf{b}$ , so  $b_{I(\mathbf{a}, \mathbf{b})} = C$  and  $\mathcal{R}$  is the set of current cooperators. Replacing every  $\pi_i(\mathbf{a})$  with  $v(\mathbf{a})$  raises all role-model payoffs and lowers all non-role-model payoffs, so by condition (2),

$$T_{\mathbf{ab}} \leq T_{\mathbf{ab}}|_{\pi_i(\mathbf{a})=v(\mathbf{a}) \text{ for all } i}$$

with strict inequality whenever  $\mathcal{R} \neq \emptyset$ . By condition (4), the right-hand side equals  $T_{\mathbf{ab}}^{\text{nd}}$ . The analogous argument with reversed inequality applies to  $D$ -gaining transitions. Hence  $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$ .

*Step 2:*  $T^{\text{nd}}$  is invariant under the  $C \leftrightarrow D$  label swap.

Let  $\sigma : S \rightarrow S$  be the map that swaps every  $C$  and  $D$ . By conditions (3) and (4), with common payoff 0, the only features of  $(\mathbf{a}, \mathbf{b})$  that enter  $T_{\mathbf{ab}}^{\text{nd}}$  are the mutation matrix and the set  $\mathcal{R}$ . Under action-invariant mutation the mutation matrix is  $\sigma$ -invariant; and  $|\mathcal{R}|$  is  $\sigma$ -invariant because the set of players currently playing  $b_{I(\mathbf{a}, \mathbf{b})}$  in  $\mathbf{a}$  has the same size as the set of players currently playing  $\sigma(b_{I(\mathbf{a}, \mathbf{b})})$  in  $\sigma(\mathbf{a})$ . Hence  $T_{\mathbf{ab}}^{\text{nd}} = T_{\sigma(\mathbf{a})\sigma(\mathbf{b})}^{\text{nd}}$  for every transition, so the stationary distribution of  $T^{\text{nd}}$  is  $\sigma$ -invariant, giving  $p_C^{(T^{\text{nd}})} = p_D^{(T^{\text{nd}})} = \frac{1}{2}$  [31].

*Step 3:*  $T^{\text{nd}}$  is stochastically monotone.

Condition (5) provides, for every  $\mathbf{a} \leq \mathbf{a}'$ , a coupling  $(\mathbf{X}, \mathbf{X}')$  on a common probability space with  $\mathbf{X} \sim T^{\text{nd}}(\mathbf{a}, \cdot)$ ,  $\mathbf{X}' \sim T^{\text{nd}}(\mathbf{a}', \cdot)$ , and  $\mathbf{X} \leq \mathbf{X}'$  almost surely. For any upper set  $U \subseteq S$ ,  $\{\mathbf{X} \in U\} \subseteq \{\mathbf{X}' \in U\}$ , so  $T^{\text{nd}}(\mathbf{a}, U) \leq T^{\text{nd}}(\mathbf{a}', U)$ , giving  $T^{\text{nd}}(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}', \cdot)$ .

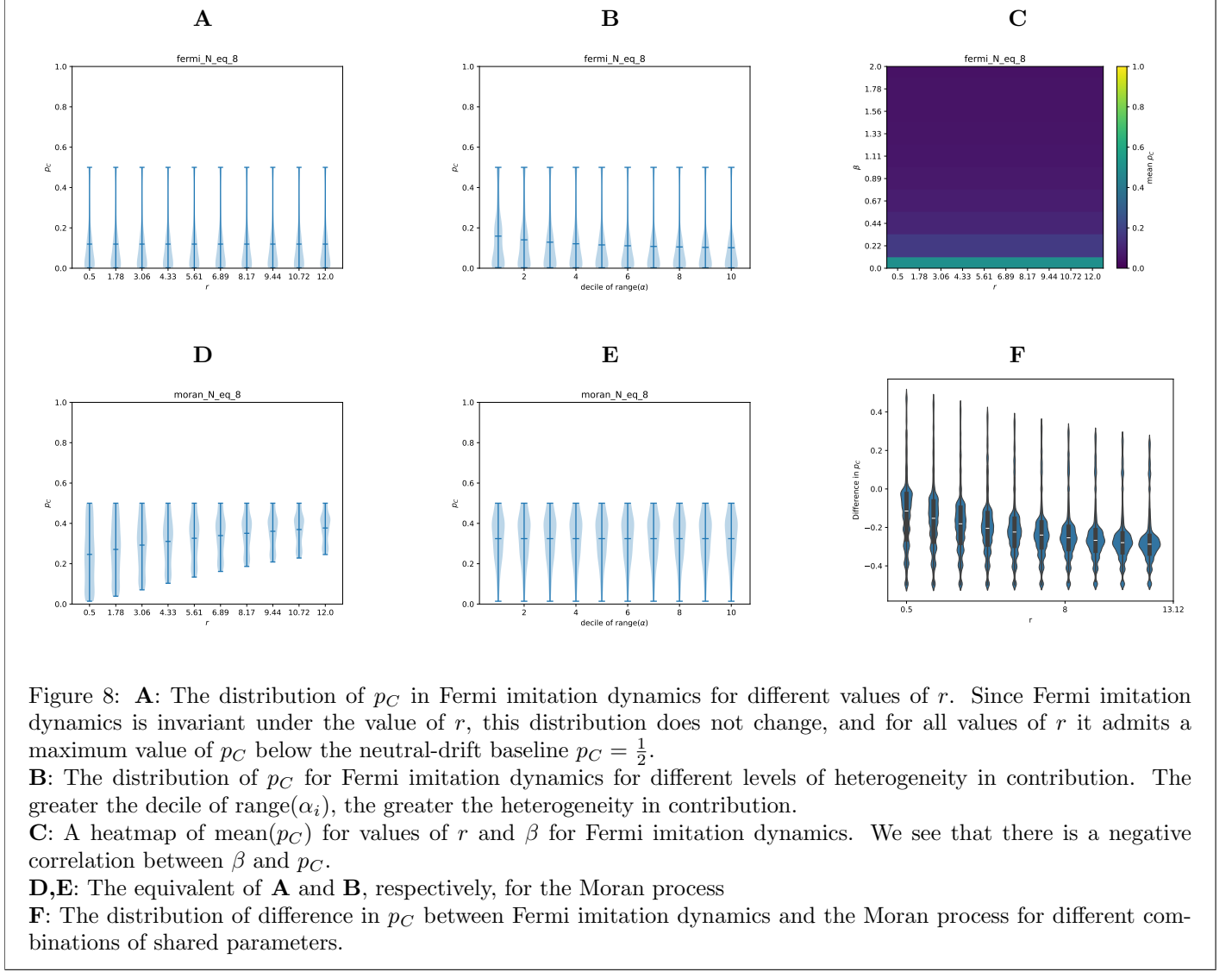


Figure 8: **A**: The distribution of  $p_C$  in Fermi imitation dynamics for different values of  $r$ . Since Fermi imitation dynamics is invariant under the value of  $r$ , this distribution does not change, and for all values of  $r$  it admits a maximum value of  $p_C$  below the neutral-drift baseline  $p_C = \frac{1}{2}$ . **B**: The distribution of  $p_C$  for Fermi imitation dynamics for different levels of heterogeneity in contribution. The greater the decile of  $\text{range}(\alpha_i)$ , the greater the heterogeneity in contribution. **C**: A heatmap of  $\text{mean}(p_C)$  for values of  $r$  and  $\beta$  for Fermi imitation dynamics. We see that there is a negative correlation between  $\beta$  and  $p_C$ . **D,E**: The equivalent of **A** and **B**, respectively, for the Moran process. **F**: The distribution of difference in  $p_C$  between Fermi imitation dynamics and the Moran process for different combinations of shared parameters.

Applying Lemma 1 to  $P = T$  and  $Q = T^{\text{nd}}$  (which satisfy (i) by Step 1 and (ii) by Step 3) yields  $\pi^T \preceq_{st} \pi^{T^{\text{nd}}}$ . Since  $n_C$  is monotone in the componentwise order,

$$p_C^{(T)} = \frac{1}{N} \mathbb{E}_T[n_C] \leq \frac{1}{N} \mathbb{E}_{T^{\text{nd}}}[n_C] = p_C^{(T^{\text{nd}})} = \frac{1}{2}.$$

□

This shows that even in a game where an action change  $D \rightarrow C$  would result in an increase in payoff for a given player, under a purely extrinsic population dynamic we cannot achieve a probability of cooperation above that of the neutral drift baseline. Purely extrinsic processes always favour the pointwise dominant strategy, regardless of which strategy provides the Nash equilibrium for the game. We now show this result in the public goods game.

### 3.4 Purely extrinsic population dynamics in the Public Goods Game

In figure ??, we show this result in the context of the public goods game. In figures **A** through **E**, we can see this limit as a hard ceiling on the value of  $p_C$  which no parameter set in our data is able to exceed. This includes parameter combinations which have high values of  $r > N$ , and so we see that despite full cooperation being the Nash equilibrium of the one shot game, the pointwise dominance of defection's fitness over that of cooperation leads to defection becoming the dominant strategy in a repeated game according to a purely extrinsic population dynamic. This not only leads to a worse average payoff for all players in the population in the steady state, but also means that when  $r > N$ , worsening moves occur with

a higher probability than changes of action types which improve a player's payoff. In other words, even when a switch  $D \rightarrow C$  would always improve a player's payoff, the player will still prefer to remain a defector under a purely extrinsic population dynamic, due to the greater fitness of a defector in a given state.

In Fermi imitation dynamics, we can also note the process' invariance under  $r$ . Figures 2 and 4 show an identical distribution of  $p_C$  for all values of  $r$ . The latter of these two figures also give that an increase in  $\beta$  will have a negative effect upon  $p_C$ , as the value  $\Delta(\pi)$  will always be negative for a  $C \rightarrow D$  transition, and positive for a  $D \rightarrow C$  transition. As the process with  $\beta = 0$  gives us neutral drift, we see our result in practice, as moving towards a less rational process increases  $p_C$  in all empirical cases.  $p_C$  in the Moran process increases with the value of  $r$ , however once again we see that it has a hard limit at  $p_C = \frac{1}{2}$ , as a given defector is always more likely to be chosen for reproduction than a given cooperator. The Moran process' ability to consider the  $r$  parameter leads it to have a higher  $p_C$  than Fermi imitation dynamics in most parameter combinations, as shown in figure 7.

## 4 Intrinsic Population Dynamics

Another classification of population dynamics is intrinsic population dynamics. In such population dynamics, a player compares their payoff to some attribute or payoff other than the payoff of other players in the population. We shall see two such population dynamics - aspiration dynamics, which compares a player's payoff to their individual baseline aspiration, and introspection dynamics, which compares a player's payoff to their counterfactual payoff when they change action type.

### 4.1 Aspiration Dynamics

Aspiration dynamics models an update rule based on how far away a player's performance in a game is from an aspirational constant fitness value:

1. A player  $i$  is chosen with probability  $\frac{1}{N}$  to reconsider their strategy  $a_{i,k}$
2. This player changes strategy with a probability  $\phi(\pi_i(\mathbf{a}) - A_i)$

Here,  $A_i$  is player  $i$ 's *aspiration* - the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is almost always defined for two actions, traditionally cooperation and defection [7], however it has been defined for games with 3 or more actions [50]. In this paper, we consider a homogeneous aspiration value  $A_i$ , and the two-action formulation of aspiration dynamics as presented in [7].

The process transitions between states according to:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N} \phi(\pi_{I(\mathbf{a}, \mathbf{b})}(\mathbf{a}) - A_{I(\mathbf{a}, \mathbf{b})}) \left(1 - \sum_{j=1}^k \mu_{I(\mathbf{a}, \mathbf{b}), j}\right) + \frac{\mu_{I(\mathbf{a}, \mathbf{b}), I(\mathbf{a}, \mathbf{b})}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a}, \mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (5)$$

The traditional two-action model of aspiration dynamics is very specific in which games it can be applied to. When aspiration is considered for three or more action types [50], a player chooses which action to change to with equal probability. Thus, players in 3-action games do not choose their strategy in the next state based on the fitness of said strategy, but instead update according to "trial and error" in order to reach their aspiration, and thus players may accept worsening moves with a high probability.

### 4.2 Introspection Dynamics

In introspection dynamics [5], individuals are selected to consider if a change of their action choice would lead to an increase in payoff. The new action is selected with a probability that depends on the potential increase and the selection intensity  $\beta$ :

1. A player  $i$  is chosen with probability  $\frac{1}{N}$  to reconsider their strategy  $a_{i,k}$
2. Player  $i$  chooses another possible strategy,  $a_{i,l}$ , from the set of all possible strategies.

3. The chosen player replaces their strategy with probability  $p_{a_{I(\mathbf{a},\mathbf{b})} \rightarrow b_{I(\mathbf{a},\mathbf{b})}}(a_{-I(\mathbf{a},\mathbf{b})}) = \phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$ , where  $\Delta\pi_{I(\mathbf{a},\mathbf{b})} = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$  is the difference between the fitness of the player's current strategy and the possible fitness of the new strategy.

This process is well suited to a heterogeneous population. This is because in some cases using imitation dynamics, a player may copy a strategy due to its fitness when played by another individual in the population, despite the fact that such a strategy actually has a lower payoff for the focal player than their current strategy. In introspection dynamics, players do not consider the fitness of other individuals, and so cannot be fooled by high fitness strategies with attributes different to that of the focal player.

The process transitions between states according to:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N(m_j - 1)} \phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})}) \left(1 - \sum_{j=1}^k \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a}, \mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (6)$$

where  $m_j$  is the number of actions available to individual  $j$ . Since we assume that all individuals have the same action set with size  $k$ , we can replace  $m_j - 1$  with  $k - 1$ . It has been shown [?] that in state-independent games, such as the public goods game and certain forms of the prisoners dilemma, introspection dynamics can be modelled as  $N$  individual markov chains, where each player changes action independently of the action types of the rest of the state. Thus,  $p_C$  breaks down into the mean  $p_i$ , the probability that player  $i$  cooperates. This makes introspection dynamics an unrealistic model of the evolution of cooperation in many common abstractions of real-world interactions, as players do not consider the current success of a strategy when choosing to accept it, but only consider the fitness of a strategy in the next step after adoption.

Introspection dynamics can be seen to follow a similar logic to classical game theory, where a well informed and rational actor maximises their own payoff. We can contrast this with purely extrinsic population dynamics, which are the traditional method of modelling populations in evolutionary games, where actors are not typically well-informed and choose their next strategy according to fitnesses in the current population, without knowledge of the impact of their transition on their fitness. Introspection dynamics allows a population to evolve by choosing new action types with full knowledge of the impact of their decision in the following one-shot game when playing a repeated game. We see below the impact which this has on a public goods game.

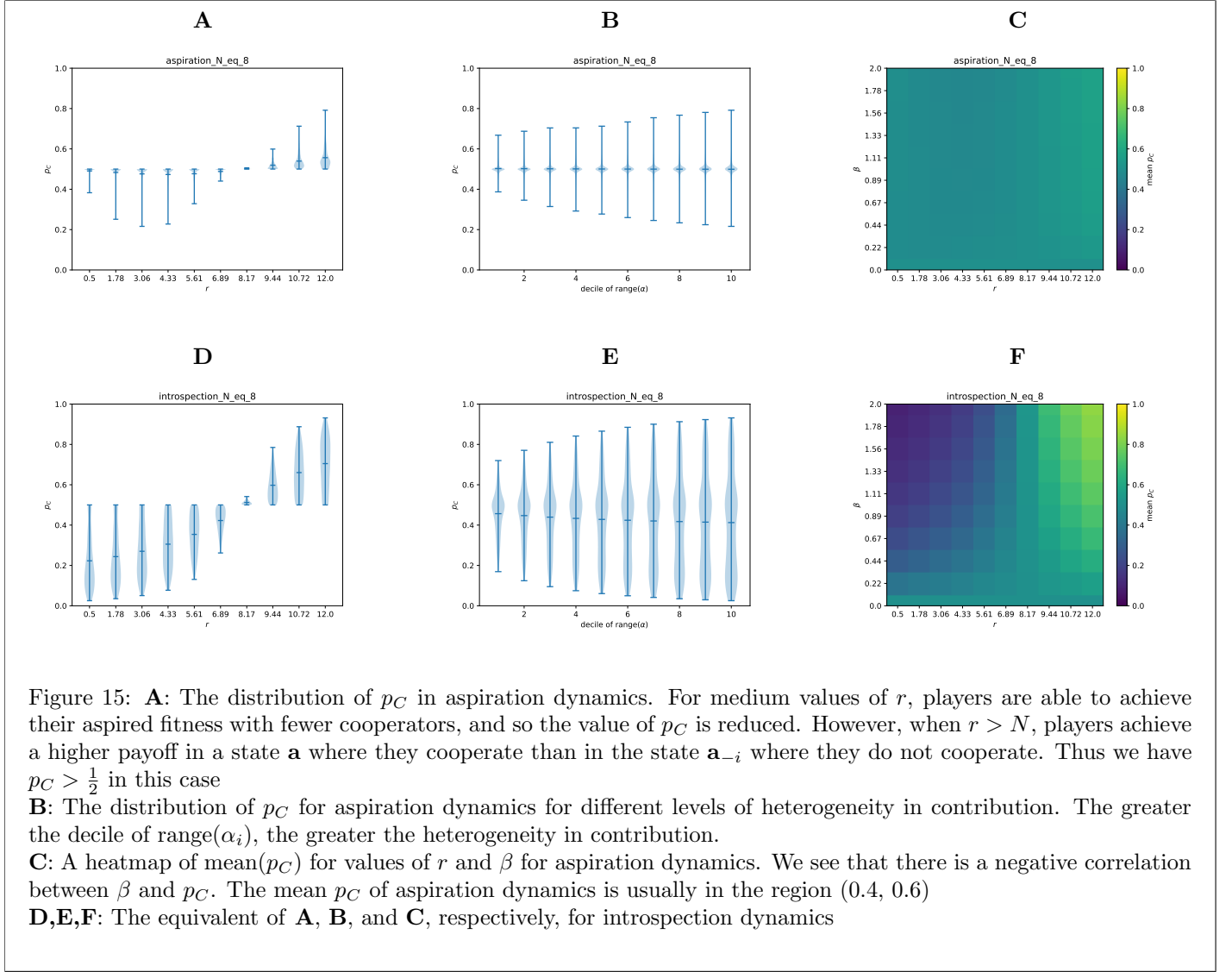
### 4.3 Intrinsic population dynamics in a public goods game

In figure ??, we see that intrinsic population dynamics are able to exceed the ceiling on  $p_C$  which is imposed on purely extrinsic population dynamics. In the case of aspiration dynamics,  $p_C$  does not reach excessively low values as in the other processes we have discussed. This is because for all players to make at least some amount of profit, at least one player must be cooperating at all times. If no players cooperate, then no player will reach their target payoff threshold. Although the ceiling is not exceeded if  $r < N$  (both shown in figure 9 and noted in [7]), we must still have multiple players cooperate in order to allow all players to reach the cooperation threshold, and the furthest above the aspiration threshold is reached when many players cooperate (although in some levels of heterogeneity, it may be that a certain number of players will defect more often in the steady state due to the large contributions needed to be made). The ceiling is exceeded at the threshold  $r > N$ , as for all players,  $\pi_i(\mathbf{a}) - A$  is greater when cooperating than defecting in a given state  $\mathbf{a}_{-i}$ , and thus the transition  $D \rightarrow C$  will always occur with a higher probability than the reverse. For intermediate values of  $r$ , aspiration dynamics can achieve it's lowest  $p_C$ , as players are able to achieve their aspiration with fewer cooperators, but still benefit from free-riding. Figure 11 shows that the mean value of  $p_C$  remains fairly steady for the values of  $r$  and  $\beta$ . The aspiration level is the biggest indicator of cooperation in the population in a linear public goods game - it has been shown previously that for a high aspiration level, cooperation emerges at a level near the neutral-drift baseline, but for a low aspiration level defectors dominate the population [7].

In the case of introspection dynamics, we can see that the ceiling is once again breached at exactly the threshold  $r = N$ .

**Proposition 1** ( $r = N$  threshold). *For any heterogeneous population with  $\alpha_i > 0$  for all  $i$ , the introspective evaluation step of Introspection Dynamics favours cooperation if and only if  $r > N$ .*

*Proof.* Individual  $i$  evaluates switching from defection to cooperation by comparing their payoff  $\pi_i(\mathbf{a})$  in the current state to the payoff  $\pi_i(\mathbf{b})$  they would receive in the state where they cooperate and all others remain unchanged. Using the



convention  $\Delta\pi_i = \pi_i(\mathbf{a}) - \pi_i(\mathbf{b})$ ,

$$\Delta\pi_i = \alpha_i - \frac{r\alpha_i}{N} = \alpha_i \left(1 - \frac{r}{N}\right).$$

Since  $\alpha_i > 0$ , this is negative if and only if  $r > N$ . Since  $\phi$  is a decreasing function,  $\phi(\Delta\pi_i) > \frac{1}{2}$  precisely when  $\Delta\pi_i < 0$ , i.e. when  $r > N$ , and  $\phi(\Delta\pi_i) < \frac{1}{2}$  when  $r < N$ .  $\square$

When  $r < N$ , the introspective step will always favour defection. This behaviour gives us a system where under introspection dynamics, players are able to choose the strategy which will maximise their own fitness at any given time step. The ability for this process to recognise the impact of their transition leads to a realistic model of real-world decision making, and a suitable population dynamic for a heterogeneous population. However, as introspection dynamics can become independent of the rest of the population in a state-independent game, it cannot always fully model learning in a real world context without making assumptions which may not reflect how individuals make decisions in the real world. Figure 14 shows this further - a stronger choice intensity favours cooperation if a switch  $C \rightarrow D$  favours cooperation, and favours defection if the reverse is true. While this is the clear rational behaviour, it is not true for Fermi imitation dynamics, as shown in figure 4.

## 5 Introspective Imitation Dynamics

We introduce our novel population dynamic *introspective imitation dynamics*. This population dynamic combines the introspective step of introspection dynamics with the fitness-proportional selection of the Moran process, in order that each individual takes into account both the fitness of strategies in the current state, and the counterfactual payoff obtained when accepting a new strategy. This models both extrinsic learning and intrinsic learning. We define the introspective imitation process as follows:

1. A focal player is chosen at random to reconsider their strategy
2. Another player  $a_j$  in the population is chosen with a probability proportional to their fitness within the population to have their strategy considered, as in the Moran process.
3. The focal player changes their strategy with probability  $\phi(\Delta(f_{I(\mathbf{a}, \mathbf{b})}))$ , as in introspection dynamics

We then define the transition matrix  $T$  of an introspective imitation process as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N} \frac{\sum_{a_j=b_{I(\mathbf{a}, \mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})} \phi(\Delta(f_{I(\mathbf{a}, \mathbf{b})})) \left(1 - \sum_{j=1}^k \mu_{I(\mathbf{a}, \mathbf{b}), j}\right) + \frac{\mu_{I(\mathbf{a}, \mathbf{b}), b_{I(\mathbf{a}, \mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (7)$$

This models a behaviour where individuals learn from each other, but do not blindly trust that the strategy will work for themselves solely because it worked for another player in the population. Instead, players look to the fitness of others in the population to see what action they *could* play, but judge based on their own fitness to decide what action they *should* play. This means that introspective imitation dynamics is well-suited to a heterogeneous population, but does not ignore the success of strategies in the rest of the population.

We see that this dynamic does not break down into individual Markov chains like introspection dynamics due to the extrinsic step, however it is shown in figure ?? able to exceed the extrinsic population ceiling under the right conditions due to the intrinsic step. It is also defined for any number of strategies, and will consider each strategy proportional to both its fitness within the state and the counterfactual payoff which the focal player is able to obtain by adopting the new action type. Finally, as the value of  $\Delta(\pi)$  is cross-state as in introspection dynamics, introspective imitation dynamics does not cancel out certain parameters in all comparisons, as in Fermi imitation dynamics.

Introspective imitation dynamics is also that it considers both a choice intensity  $\beta$  and a selection intensity  $\epsilon$ . This means that the parameters  $\epsilon$  and  $\beta$  are both tunable to emphasise either the extrinsic or intrinsic step of the dynamic.

### 5.1 introspective imitation dynamics in the public goods game

We see in figure ?? that in the same manner as the purely intrinsic population dynamics, introspective imitation dynamics is able to exceed the  $p_C = \frac{1}{2}$  ceiling which is imposed by purely extrinsic population dynamics. Despite the extrinsic step in the dynamic, the intrinsic step is sufficient to enable the dynamic to produce high values of  $p_C$  under the right parameters. We can see in figure ?? that this occurs exactly at the threshold  $r = N$ : the same threshold that we see  $\phi(\Delta(\pi))$  begin to favour cooperation in introspection dynamics. Before this threshold, we actually see in figure ?? that introspective imitation dynamics favours defection more severely than in the standard Moran process, as both steps suppress the probability of a  $D \rightarrow C$  transition. We can see, therefore, that the intrinsic step of introspective imitation dynamics and introspection dynamics will favour the strategy which admits a Nash equilibrium in the one-shot public goods game (i.e, cooperation if the Nash equilibrium is for all players to cooperate, and defection if it is for all players to defect).

This does not necessarily imply that introspective imitation dynamics admits a value of  $p_C$  which favours cooperation in line with  $\phi$  when  $r > N$ , as certain choices of  $\epsilon$  and  $\beta$  may cause the process to emphasise the extrinsic step, which will always favour defection, over the intrinsic, as seen in figure ?. We have seen previously for introspection dynamics that a stronger choice intensity favours cooperation if  $r > N$  and favours defection if the inequality is reversed - this remains true in introspective imitation dynamics. A stronger selection intensity on the other hand always favours defection. There is no result which says, for a given pair of choice and selection intensity that favours defection, that there is no value of  $r$  which admits a  $p_C > \frac{1}{2}$  - in fact it is likely that such a value exists outside of the range of our empirical data, as the difference between a cooperator and defector in the extrinsic step becomes minimal, and the intrinsic step strongly favours cooperation.

Previously, we saw that heterogeneity had no effect on  $p_C$  in the Moran process (figure 6), and that a greater level of heterogeneity caused a lower mean and more polarised extreme values in introspection dynamics (figure ??). In introspective imitation dynamics, heterogeneity causes the mass of the distribution to approach the extremes of  $p_C$ . This

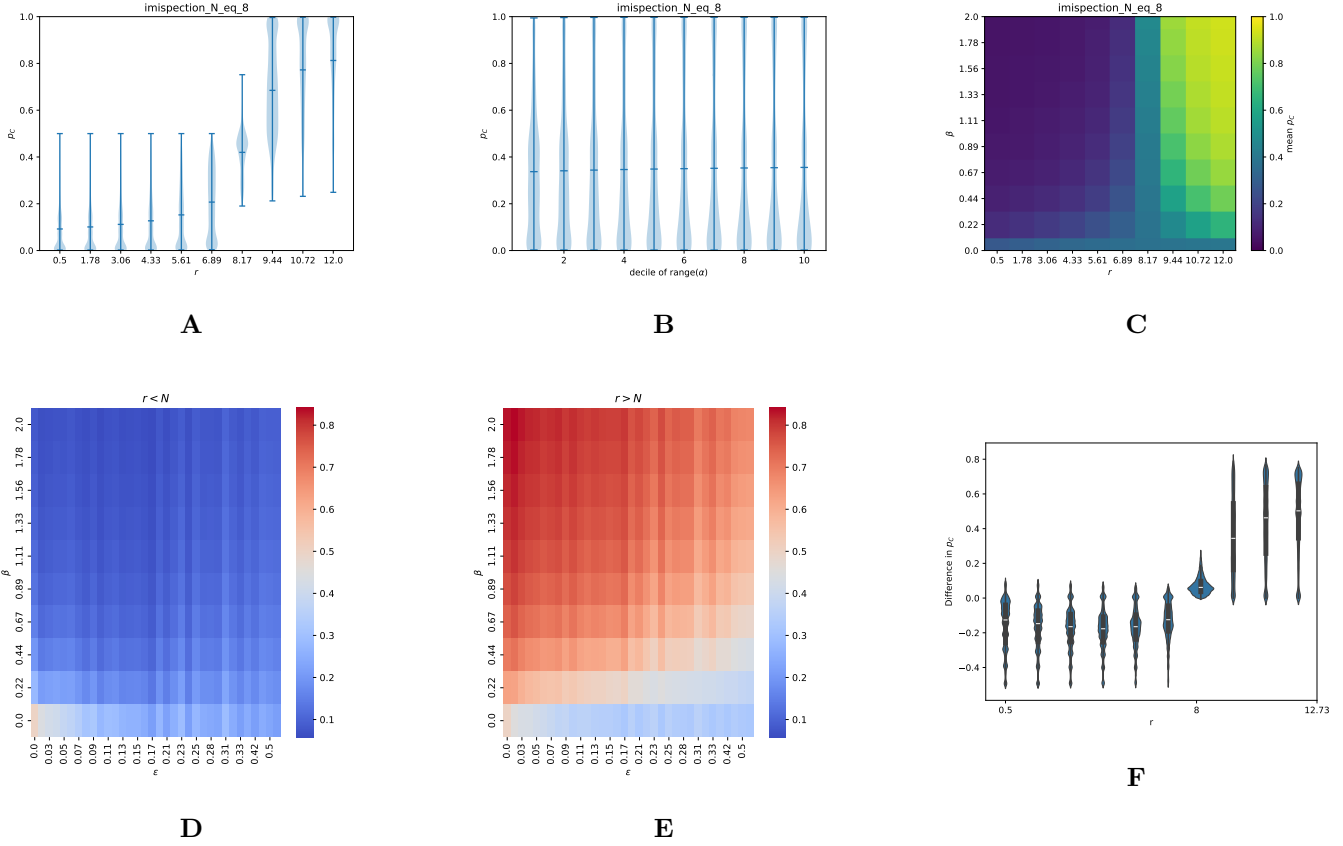


Figure 16: **A:** the distribution of  $p_C$  in introspective imitation dynamics for various values of  $r$ . **B:** the distribution of  $p_C$  in introspective imitation dynamics for various levels of heterogeneity. The greater the value of  $\text{range}(\alpha)$ , the greater the contribution heterogeneity. **C:** A heatmap of  $\text{mean}(p_C)$  in introspective imitation dynamics for varying  $r$  and  $\beta$ . We see a threshold for the effect of a stronger choice intensity at  $r = N$ . **D,E:** Heatmaps of the effect of both selection and choice intensity on  $\text{mean}(p_C)$  for  $r > N$  and  $r < N$ . **F:** The difference in  $p_C$  between introspective imitation dynamics and the Moran process for populations with shared parameters, specifically for different values of  $r$ .

effect can once again be attributed to the threshold  $r = N$ . In a very heterogeneous population, high contributors have a comparatively much worse payoff to defectors, and thus are chosen by the extrinsic step of the dynamic more often. One of two systems then appear. If  $r < N$ , high contributors are tempted with a large improvement in payoff if they defect as  $\alpha_i \left( \frac{r}{N} - 1 \right)$  is much lower. Thus, high contributors cooperate much less in a more heterogeneous population with a low  $r$ . If  $r > N$ , then the return  $\frac{r\alpha_i}{N}$  given by a high contribution is much larger in a more heterogeneous population. Thus, the intrinsic step of the population dynamic will favour cooperation much more strongly in a population with a greater level of heterogeneity if  $r > N$ . We see this polarisation once again favours the strategy which maximises the overall payoff of a given player.

## 6 Conclusion

We have introduced and analysed a framework for heterogeneous evolutionary games under five distinct population dynamics in three categories, as shown in figure 18. We have defined the categorisation of a *purely extrinsic population dynamic*, and given rigorous formulations of two widely used examples in the Moran process and Fermi imitation dynamics. We have shown that dynamics under this classification have a hard ceiling on the probability of cooperation  $p_C$  in social dilemmas where a defector *always* outperforms a cooperator in a given state. In such a game, purely extrinsic population dynamics are unable to exceed the neutral-drift baseline. We then give formulations of two intrinsic population dynamics

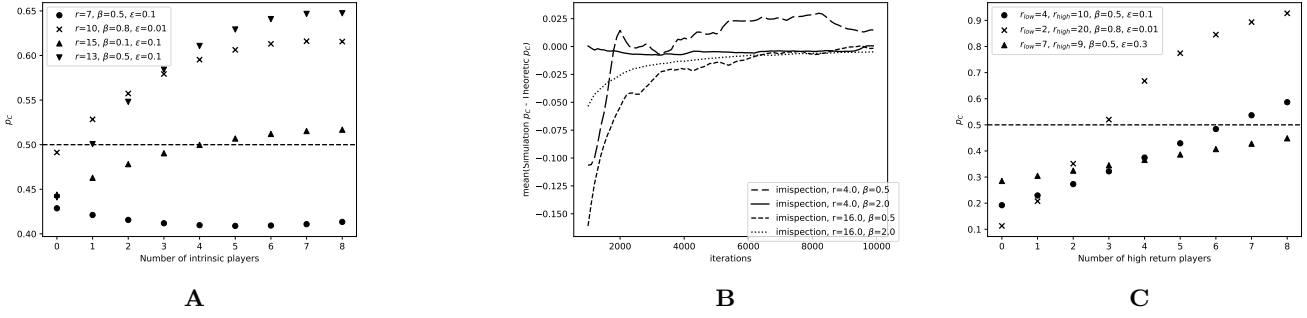


Figure 17: **A:**  $p_C$  in various parameter combinations at  $N = 8$ , varying the number of players using an intrinsic population dynamic. 0 intrinsic players gives the  $p_C$  of the Moran process, whereas 8 gives us the  $p_C$  in introspective imitation dynamics. We take a linear population with a maximum contribution of 8.

**B:** for various parameter combinations, the mean value over numerous simulations of an 8 player population playing introspective imitation dynamics converging to the theoretic  $p_C$ .

**C:**  $p_C$  for a heterogeneous  $r$  in introspective imitation dynamics. 0 high return players corresponds to a population with homogeneous return  $r_{low}$ . 8 high return players corresponds to a population with a homogeneous return  $r_{high}$ .

- aspiration dynamics in which players change strategy based on their performance in relation to a target payoff, and introspection dynamics in which players compare their current payoff with their counterfactual payoff which they could obtain by switching action type. We have also shown empirical results for the public goods game with heterogeneous contributions which confirm the ability for intrinsic population dynamics to exceed the population ceiling imposed by purely extrinsic processes. Finally, we have introduced a novel population dynamic in Introspective imitation dynamics with both an extrinsic and intrinsic step. Our empirical results confirm that this dynamic is able to overcome the ceiling of purely extrinsic population dynamics, but still allows players to take into account the performance of strategies within the current state. We establish the threshold  $r = N$  for which the intrinsic step in introspection dynamics and introspective imitation dynamics are able to overcome the neutral-drift baseline.

Further work on this may look further into the effect of hybrid population dynamics, similar to the combination of Moran and Fermi processes in [25], where extrinsic and intrinsic population dynamics are played in a mixed population. Figure ?? shows three populations where the population dynamic is treated as a heterogeneous attribute, with players learning according to either the Moran process or introspective imitation dynamics. We see that under different parameters, a different number of players may possess an intrinsic learning method in order to exceed the neutral drift baseline  $p_C$ . show that simulations converge to the theoretic value in introspective imitation dynamics across a large number of iterations in introspective imitation dynamics. Further study may make use of mixing times [24] in order to show which population dynamics converge to their theoretic  $p_C$  in the fewest iterations. Finally, the heterogeneity considered in this paper is in the player's contributions, however the threshold for introspective imitation dynamics occurs at the point  $r = N$ . Figure ?? shows that in introspective imitation dynamics with certain heterogeneous values of  $r_i$ , fewer players are required to receive a return  $r > N$  in order to exceed the cooperation threshold. Additionally, we see in both figures ?? and ?? that under certain values of  $\epsilon$  and  $\beta$ , even a population which is entirely contributing a high value of  $r$  will be unable to exceed the  $p_C$  threshold, as the extrinsic step of the process occurs with a higher rationality. The relationship between such values and the heterogeneity of  $r$  may be a focus of further study.

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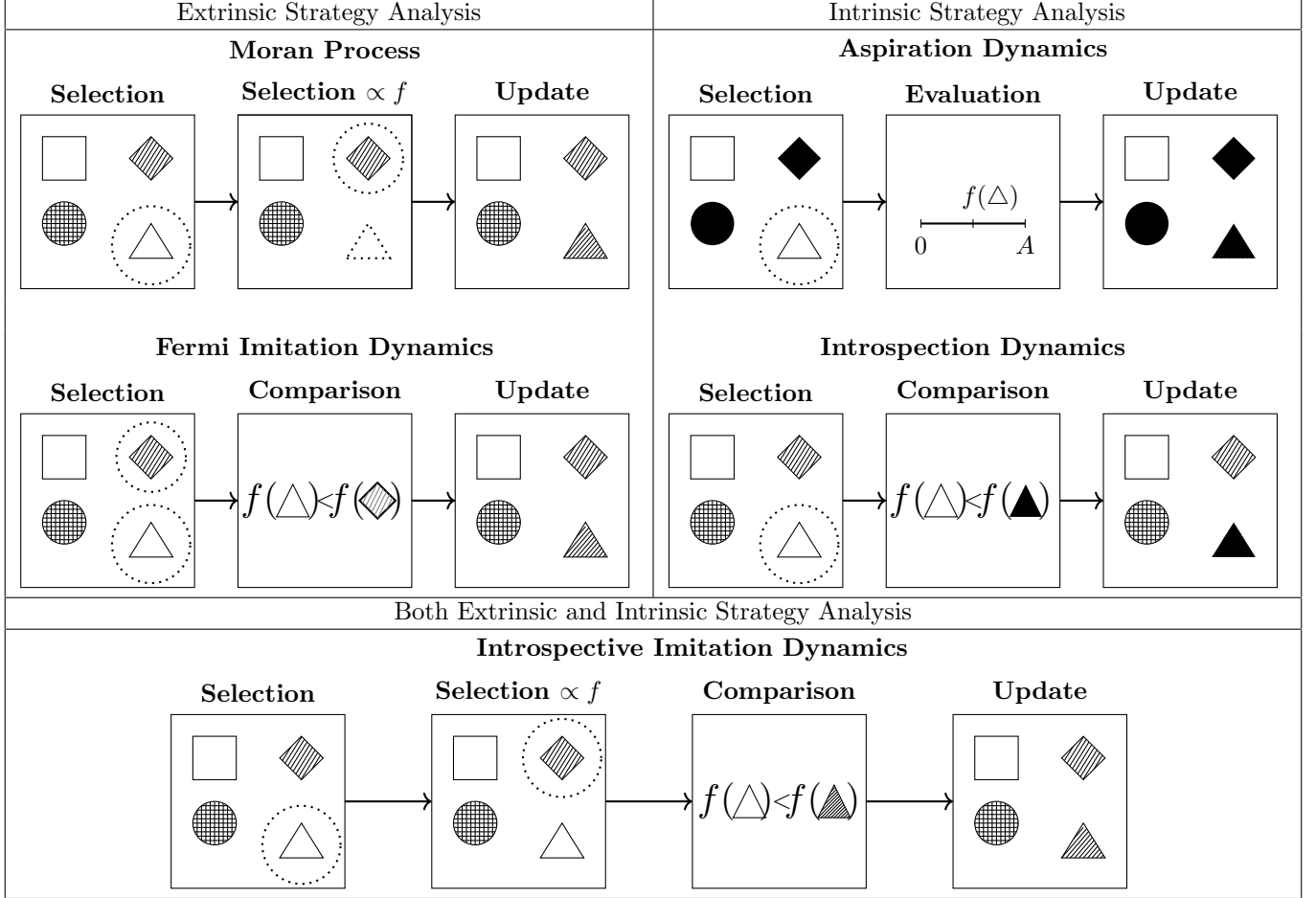


Figure 18: A comparison of different population dynamics. We assume in all cases that the triangle is chosen for replacement with a probability  $\frac{1}{N}$  and the diamond is chosen with a varying probability depending on the process. In the case of Introspection dynamics, we assume that the strategy represented by the shape being filled in is chosen with probability  $\frac{1}{k-1}$ , and for aspiration dynamics there is only one other strategy to accept.

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